SCU SEEDs Workshop

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This Talk

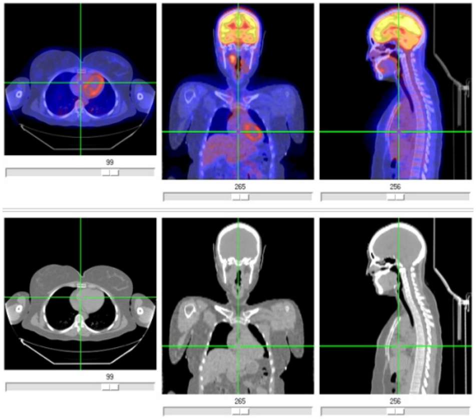
- Part I — Computing
- Part II — Computing at SCU
- Part III —Today's activity

PART I — COMPUTING

What is Computing?

- Analysis, design and development of computer systems
- It is not just about programming
- It teaches you how to think more methodically and how to solve problems more effectively

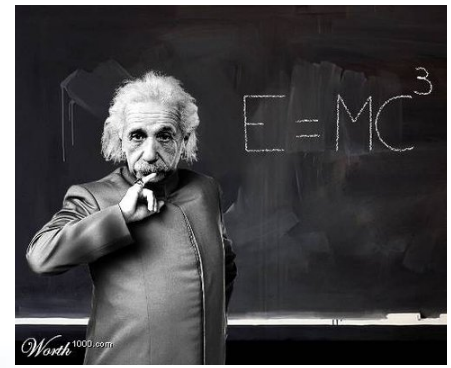
Computing is everywhere!



What is Computing?

- Computing includes a variety of fields:
  - Mathematics
  - Computer science
  - Computer engineering
  - Information science
  - Electrical engineering

What is Computing?



- What is a computer professional?
  - Will I have to grow fuzzy hair?

- What does s/he do?
  - Will I have to sit in front of a computer all day?

- What kind of people will I work with?
  - Will I have to become a geek nerd?

What is Computing?

- FUN, COOL, and EXCITING
  - Cutting edge projects
  - Exciting and talented people
  - All over the world, in every sector
  - Significant impact on society and our planet

Why Study Computing?

- **Intellectually interesting**
  - Logical reasoning and mathematical thinking
  - Possible workings of the human mind

Why Study Computing?

- **Computing supports and links to most other areas of study**
    - Computing and neuroscientists – the brain
    - Computing and Biologists – Genome
    - Computing and Meteorologists – weather prediction
  - Future scientists require basic knowledge of Computing

Why Study Computing?

- **Computing teaches problem solving**
  - Decomposition, abstraction, modular design
  - Analysis and design are carefully reviewed
  - Always new methods being investigated

Why Study Computing?

- **Computing builds teamwork and leadership skills**
  - Plan, organize, control, lead complex projects
  - Learn to deal with mix of talents
  - Estimate and deal with risk

Why Study Computing?

- **Computing develops life-long learning skills ...** ***"Change is the only constant"***

– Promotes learning to learn

"if GM had kept up with the technology like the computer industry has, we would all be driving $25.00 cars that got 1,000 miles to the gallon" – Bill Gates

– Exponential growth makes many predictions look foolish

False Predictions

- "*I think there is a world market for maybe five computers*" -- Thomas J. Watson, founder and Chairman of IBM, 1943.

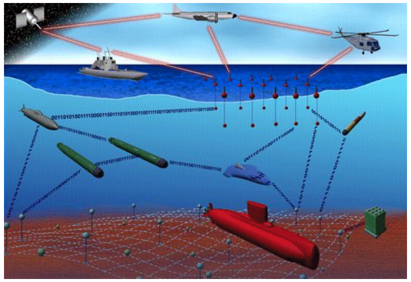
- "*Computers in the future may weigh no more than 1.5 tons*" -- *Popular Science*, 1949.

- "*640K ought to be enough for anybody*" -- Bill Gates, 1981.

Future Applications

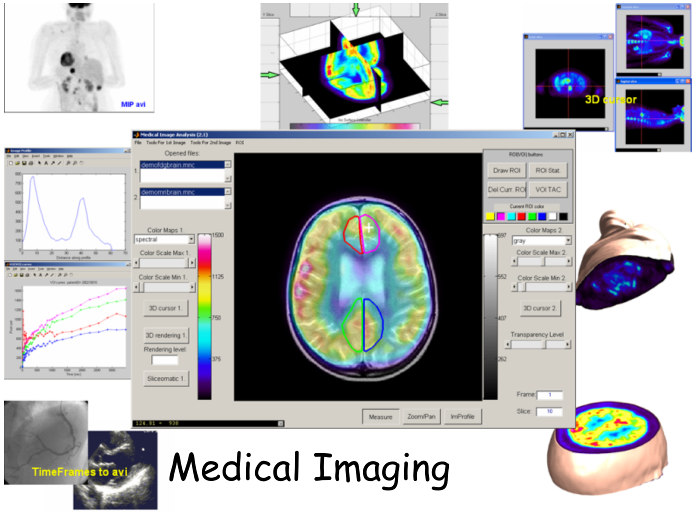
Self-driving car

Personalized Healthcare



Transforming the nation's defense

Internet of Things



Medical Imaging

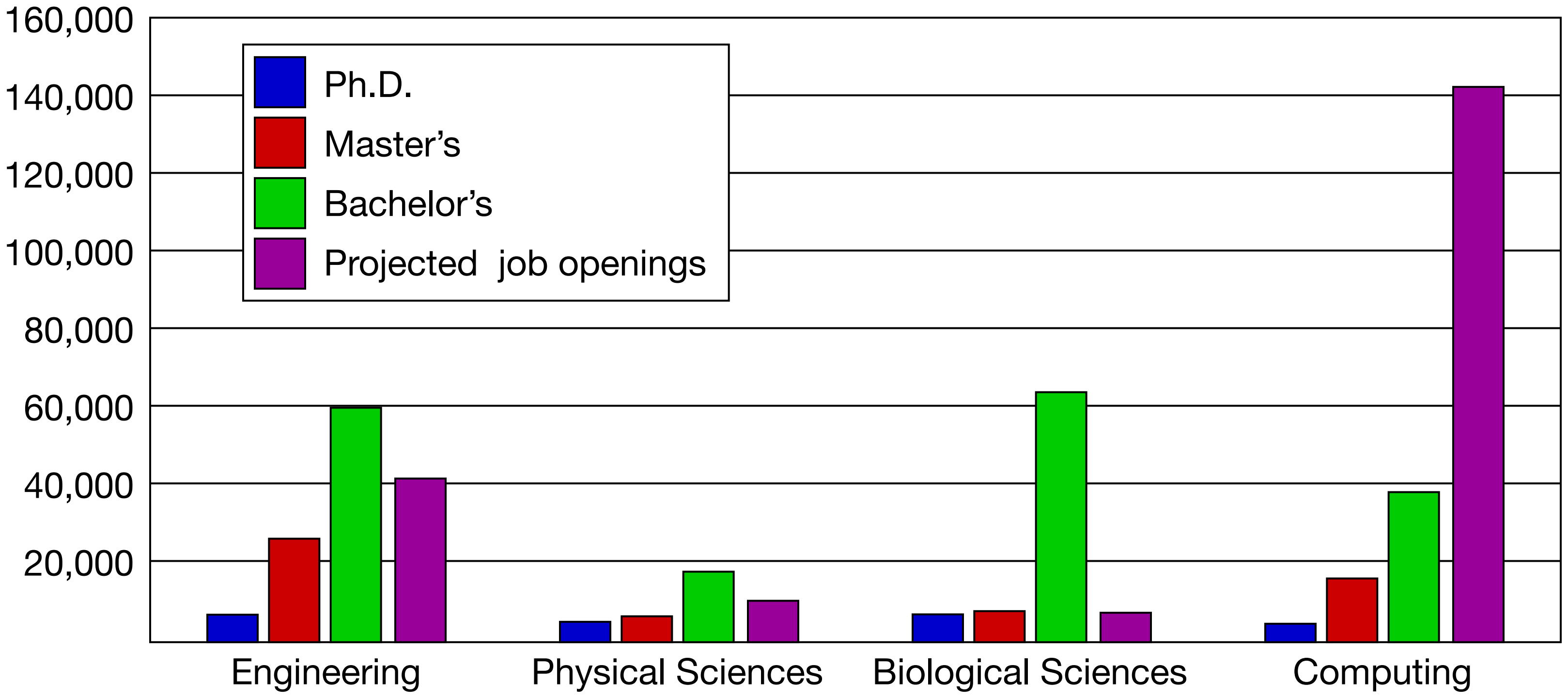
Comp. Science & Engineering

- Computer science
  - Often more mathematical
  - Computability theory
  - Algorithmic complexity
- Computer engineering
  - Often more hardware-oriented
  - Image and signal processing
  - Computer graphics

Career Opportunities

- System architect
- Network engineer
- Computer architect
- Software engineer
- Security specialist
- Game designer
- Test engineer
- Entrepreneur, musician, athlete, and more

Degree Production *vs.* Job Openings



Sources: Adapted from a presentation by John Sargent, Senior Policy Analyst, Department of Commerce, at the CRA Computing Research Summit, http://www.cra.org/govaffairs/content.php?cid=22.

Be Creative!

- Computing is the only tech field in which you can create a product from scratch and commercialize it independently

- Computing is the only tech field in which you can easily come up with your own personal solutions.

PART II — COMPUTING AT SCU

Computing Degrees at SCU

- Undergraduate degrees
  - Computer science and engineering (CSE)
  - Web design and engineering (WDE)
  - Mathematics and computer science
- Graduate degrees
  - Computer science and engineering
  - Software engineering
- 5-year Master’s program

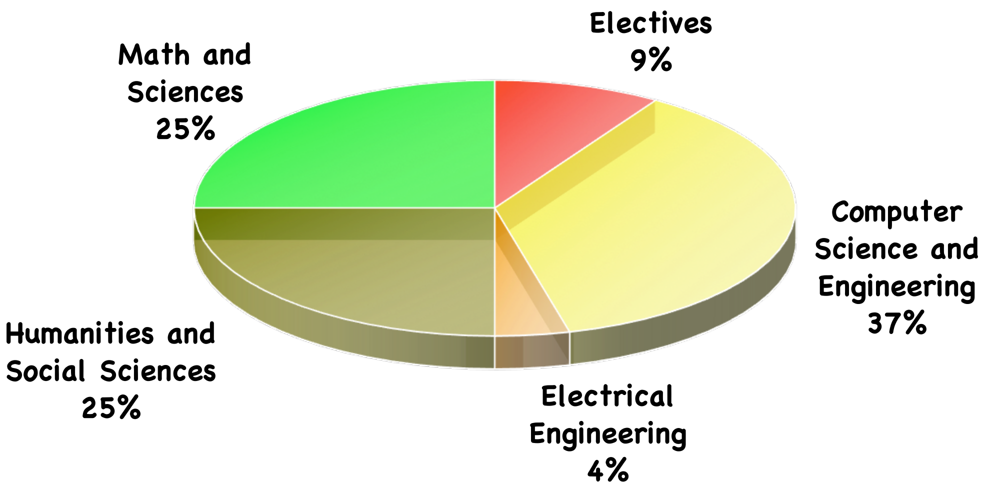
Undergraduate CSE

- Combination of computer science and computer engineering
- Focuses on theoretical and practical aspects of computing
- Design and construction of both hardware and software systems
  - Computer networks, operating systems, algorithms, compilers, software engineering, embedded programming, Web programming, robotics, 3D animation

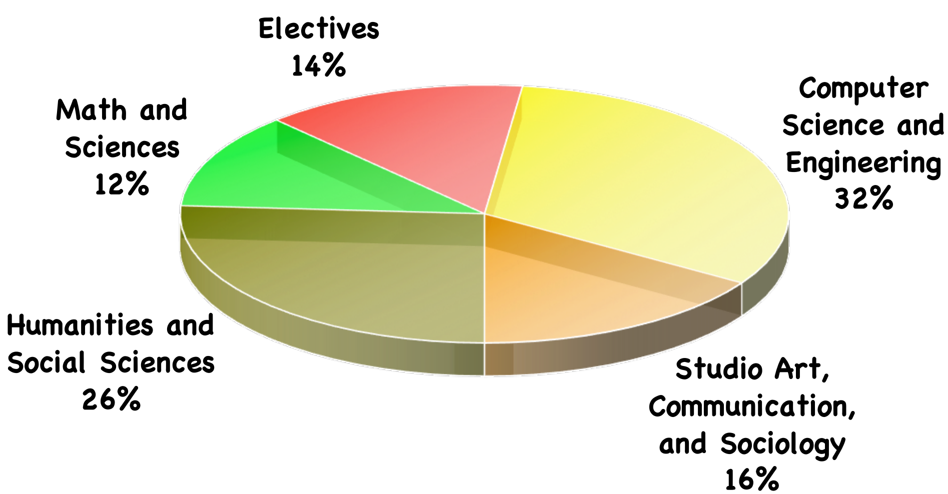
Undergraduate WDE

- New major started in 2009
  - One of the first such programs in the country
- Combines computing with other disciplines:
  - Graphic arts
  - Communication
  - Sociology
- What will these specialized graduates do?
  - Improve Web infrastructure
  - Develop interactive, multimedia content
  - Analyze the huge amount of information on the Web (Big data)
  - Understand the societal impact of the Web

Coursework



Computer Science and Engineering



Web Design and Engineering

Where Will You Work?

- Recent graduates went to work for:
  - Cisco, Apple, Microsoft, IBM, Google, Facebook, Groupon, Amazon, Anritsu, F5 Networks
  - Starting salary range: $70K–$100K
- Recent graduates also continued their education:
  - Ph.D program at Berkeley, UCSD, etc.
  - M.S. programs at SCU, CMU, Stanford, etc.

PART III — TODAY’S ACTIVITY

PROGRAMMING IN PHP

What Is a Program?

- **Program**
  - A set of instructions that tells the computer what to do.
  - Used to solve a specific problem.
- **Algorithm**
  - A particular sequence of operations.
  - Written using a computer language (such as Java, C, C++, Python, orPHP) to create a program.

What is PHP?

- PHP is an acronym for "Hypertext Preprocessor"
- PHP is a widely-used, free, open source scripting language
- PHP is a server scripting language, and a powerful tool for making dynamic and interactive Web pages.
- PHP scripts are executed on the server
- PHP is free to download and use

Source: http://w3schools.com/php

PHP

- **PHP is an amazing and popular language!**
- It is powerful enough to be at the core of the biggest blogging system on the web (WordPress)!
- It is deep enough to run the largest social network (Facebook)!
- It is also easy enough to be a beginner's first server side language!

Source: http://w3schools.com/php

What is a PHP File?

- PHP files can contain text, HTML, CSS, JavaScript, and PHP code
- PHP code are executed on the server, and the result is returned to the browser as plain HTML
- PHP files have extension ".php"

Source: http://w3schools.com/php

Basic PHP Syntax

- A PHP script starts with **<?php** and ends with **?>**

- Example:

```
<?php
// PHP code goes here
?>
```

Source: http://w3schools.com/php

Basic PHP Syntax

- A PHP file normally contains HTML tags, and some PHP scripting code.
- Example:

```
<html>
<body>

<h1>My first PHP page</h1>

<?php
echo "Hello World!";
?>

</body>
</html>
```

Source: http://w3schools.com/php

Programming

General goal of a program

Receive an input → Produce an output

Fundamental Components

- Variables and constants
- Statements

Statements

- Specifies an action to be performed.
- Each statement is ended with a semicolon.
- Words are usually separated by a single space.
- There should not be a space *within* a word.
- Examples:

```
echo "Hello World!";
$x = 5 + 15;
echo $x;
$color = "red";
echo "My car is " . $color . "<br>";
```

Programming

Variables

- Have a unique name
- Receive initial values
- Change values as the code executes
- Holds final values

Basic Statements, end with a semicolon

- Assignments
- Calls

Programming

Flow Statements

- Sequence
  - Statements happen in the order defined by the program
- Condition
  - Depending on a condition, define the next step
- Counting loop
  - Repeat a set of statements a number of times
- Conditional loop
  - Repeat a set of statements while a condition is true

PHP

Variables

- Names start with a $ sign followed by an alphanumeric character or an underscore

Assignments

```
$x = 1;
$x = $y + $z;
$x = $x + 1;
echo $x;
```

PHP

Condition

```
if (condition)
{
	statements
}
else
{
	statements
}
```

Condition

==, >, <, >=, <=, !=

PHP

Condition, example

```
if ($x == 0)
{
	$x++;
}
else
{
	$x--;
}
```

PHP

Repetition

```
while (condition)
{
	statements
}
```

Condition

 ==, >, <, >=, <=, !=

PHP

Repetition, example

```
$x = 1;
while ($x <= 5)
{
	echo $x;
	$x++;
}
```

Project

```
<html>
<body style = 'color:red ; background-color:blue'>

<?php

$name = ”Angela";

$i = 1;
while ($i <= 5)
{
		echo "<big>";
		echo  $name;
		echo "<br>";
		$i++;
}

?>

</body>
</html>
```

Project

Use phpfiddle.org

Get the code from
www.cse.scu.edu/~amusurlian/big

Project

Tasks

- Change the name and colors
- Change the loop to have the first name have a different color
  echo "<p style='color:red'>";
- Change the loop to use two colors
  - First half/Second Half
  - Even/odd
    if ($i % 2 == 0)
        even
    else
        odd
- Add another loop to write smaller and smaller

Summary

- Part I — Computing
- Part II — Computing at SCU
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- Computing fields are a lot of fun!

Thanks!

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